

water heater suitably. Normally, the circuit remains idle and no current flows through it.

Three different types of circuits are described here. The circuit in Fig. 1 shows over-heating status of the pipe through LED1. As soon as the temperature of water pipe rises above the preset limit, the N/O contact of the thermostat closes and current flows through LED1. The LED1 glows to indicate that the temperature of the

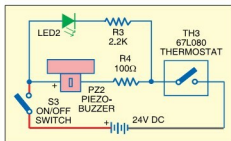


Fig. 3: Circuit for audio-visual indication

water pipe has exceeded.

The circuit in Fig. 2 gives aural in-

dication of over-heated pipe through piezobuzzer PZ1. When the temperature of the pipe goes above the set limit, the N/O contact of the thermostat closes to sound the piezobuzzer.

The circuit in Fig. 3 gives audio-visual indication through piezobuzzer PZ2 and LED2. When the temperature of the pipe goes above the set limit, the N/O contact of the thermostat closes to light up LED2 and sound piezobuzzer PZ2. ●

SIMPLE KEY-HOLE LIGHTING DEVICE

■ T.A. BABU

When opening the door of your room in the dark, you may have to spend a few

minutes searching for the key-hole. This simple circuit is an extracted circuit board from a discarded quartz timepiece. Most hobbyists might be having one such unit in their junk box. First, disconnect the coil and its fittings, noting the pads that it is connected to. The other components can then be connected across the pads as shown in the circuit diagrams. There are two circuits: manual (shown in Fig. 1) and automatic (shown in Fig. 2).

that makes it easy for you to open the door in the dark. An ultra-bright red LED in this circuit flashes rather than lighting continuously.

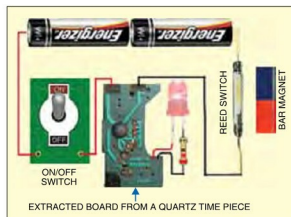


Fig. 1: Simple key-hole lighting device (manual)

minutes searching for the key-hole. Here is the circuit of a lighting device

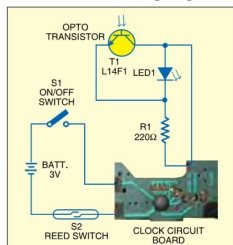


Fig. 2: Circuit for key-hole lighting device (automatic)

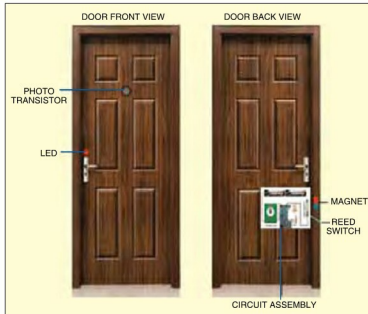


Fig. 3: Door assembly

(when there is sufficient light). An on/off switch is provided to switch off the unit when it is not required. This circuit design, if properly mounted, will decorate your door at an amazing low assembling cost. After proper fabrication, fix the complete gadget above your door lock as shown in Fig. 3. ●